

SMARTEDIT: A HUMAN-IN-THE LOOP AI FRAMEWORK FOR MULTI-FOOTAGE VIDEO EDITING

SUDHI BABU
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Mini Project

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Outline

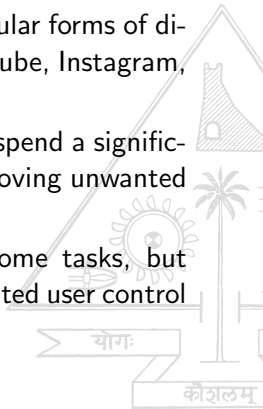
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Introduction

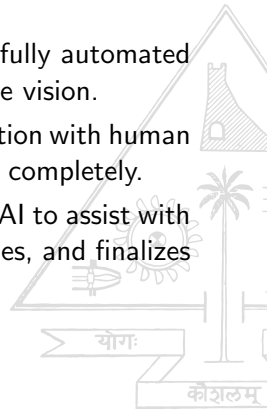
- Video content has become one of the most popular forms of digital communication on platforms such as YouTube, Instagram, and in professional media production.
- Creating high-quality videos requires editors to spend a significant amount of time reviewing raw footage, removing unwanted clips, trimming silence, and arranging scenes.
- Existing AI-powered editing tools automate some tasks, but they often produce generic edits and provide limited user control over the final output.





Introduction

- Fully manual editing is time-consuming, while fully automated editing often fails to match the editor's creative vision.
- Editors need a system that combines AI automation with human decision-making instead of replacing the editor completely.
- A Human-in-the-Loop (HITL) approach allows AI to assist with repetitive tasks while the editor reviews, modifies, and finalizes the editing decisions.

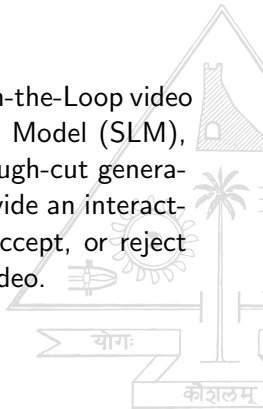




Introduction

Objective:

- To develop **SmartEdit**, an AI-assisted Human-in-the-Loop video editing framework that uses a Small Language Model (SLM), OpenCV, Librosa, and FFmpeg to automate rough-cut generation, detect shaky footage and silence, and provide an interactive timeline where users can review, modify, accept, or reject AI-generated edits before exporting the final video.





Literature Review

T. Tilekbay, A. Kim, D. E. Park, and J. A. Landay, “ExpressEdit: A Multimodal System for Video Editing with Natural Language and Sketching,” *Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2024.

- Proposed a multimodal video editing system using natural language and sketches.
- Made video editing easier through intuitive user interactions.
- **Observation:** Focuses on editing interactions only. It does not support AI-assisted rough-cut generation, shaky-footage detection, silence removal, or Human-in-the-Loop (HITL) editing as proposed in SmartEdit.



Literature Review

Z. Chen, K. Liang, H. Wang, et al., “EditBoard: A Comprehensive Evaluation Benchmark for Text-based Video Editing,” *arXiv preprint arXiv:2402.09257*, 2025.

- Introduced **EditBoard**, a comprehensive benchmark for evaluating AI-based text-driven video editing models.
- Provided standardized datasets, evaluation tasks, and performance metrics to compare different video editing approaches fairly.
- Helped researchers analyze the strengths and weaknesses of text-based video editing systems using a common evaluation framework.
- **Observation:** The work focuses on evaluating AI video editing models rather than building an interactive editing system. It does not support Human-in-the-Loop (HITL) editing, rough-cut



Literature Review

K. Shekar, K. Kasi Viswanadh, V. Srujan Varma, K. Veda Sri, S. Aditya, and S. Sahoo, "AI-Based Intelligent Video Editing System," *2025 International Conference on Advancements in Smart, Secure and Intelligent Computing (ASSIC)*, 2025.

- Proposed an AI-assisted video editing system that automates video editing tasks such as scene selection and clip organization using artificial intelligence techniques.
- Reduced manual editing effort by automatically identifying important video segments and generating edited video content.
- Demonstrated the effectiveness of AI in improving editing speed and simplifying the video production process.
- **Observation:** The system focuses mainly on automated editing and provides limited user interaction. It does not support Human-in-the-Loop (HITL) editing, natural language prompt



Literature Review

S. Tu, Q. Dai, Z. Zhang, S. Xie, Z.-Q. Cheng, C. Luo, X. Han, Z. Wu, and Y.-G. Jiang, "Paper Title," *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.

- Proposed an advanced AI-based video editing framework using computer vision techniques for understanding video content and automating editing tasks.
- Applied deep learning models to analyze video scenes, detect important events, and improve the quality of AI-generated video edits.
- Demonstrated the effectiveness of computer vision in reducing manual editing effort and enhancing editing efficiency.
- **Observation:** The framework mainly focuses on automated video editing and visual content analysis. It does not support Human-in-the-Loop (HITL) editing, natural language prompt-



Literature Review

B. Li, Y. Wu, Y. Lu, J. Yu, L. Tang, J. Cao, W. Zhu, Y. Sun, J. Wu, and W. Zhu, "VEU-Bench: Towards Comprehensive Understanding of Video Editing," *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

- Introduced **VEU-Bench**, a comprehensive benchmark for evaluating AI models on a wide range of video editing tasks.
- Provided standardized datasets, evaluation metrics, and benchmark tasks to measure the video understanding and editing capabilities of AI systems.
- Enabled fair comparison of different video editing models by assessing their performance across multiple editing scenarios.
- **Observation:** The work focuses on evaluating AI video editing models rather than developing an interactive editing applic-



Literature Review

Author	Approach	Brief Observation
Tilekbay et al. (2024)	Multimodal video editing	Supports video editing using natural language and sketches, but lacks AI-assisted rough-cut generation and HITL editing.
Chen et al. (2025)	Benchmark for text-based video editing	Provides evaluation datasets and metrics, but does not develop an interactive video editing system.
Shekar et al. (2025)	AI-based video editing	Automates scene selection and editing, but offers limited user control and no HITL workflow.
Tu et al. (2025)	Computer vision-based video editing	Uses computer vision for intelligent video editing, but focuses mainly on automation rather than user-guided editing.
Li et al. (2025)	VEU-Bench evalu-	Introduces a benchmark for eval-



Existing System

- **Manual video editing:** Editors must review raw footage, remove unwanted clips, trim silences, and arrange videos manually, making the process time-consuming.
- **AI-powered editing tools:** Existing tools automate tasks such as scene detection, silence removal, and clip selection, but provide limited user control over editing decisions.
- **Prompt-based editing systems:** Some recent systems allow editing through natural language, but they mainly focus on applying user commands rather than generating intelligent rough cuts.
- Most existing solutions operate as fully automated systems, offering little opportunity for users to review, modify, or refine AI-generated edits.
- There is no unified framework that combines natural language prompt interpretation, computer vision, audio analysis, AI-assisted rough cut generation, and natural language (NLP) editing.



Proposed System

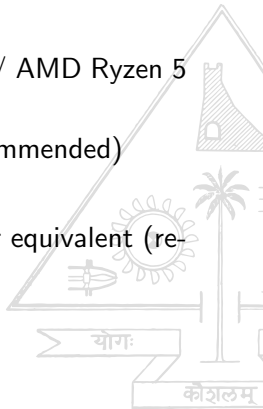
- **SmartEdit** is an AI-assisted Human-in-the-Loop (HITL) video editing system that speeds up rough-cut creation while keeping the editor in control.
- **Natural Language Prompt:** Users describe the desired editing style and video duration using simple text prompts interpreted by a Small Language Model (SLM).
- **AI Video Analysis:** OpenCV detects shaky or low-quality footage, while Librosa identifies silence for automatic audio trimming.
- **Intelligent Rough-Cut Generation:** AI automatically selects, trims, and arranges clips based on the user's editing preferences.
- **Interactive Timeline:** Editors can preview, accept, reject, rearrange, or modify AI-generated edits before exporting the final video.
- Reduces manual editing time while preserving creativity through AI-assisted video editing.



Requirements Specification

Hardware Requirements

- **Processor:** Intel Core i5 (10th Gen or higher) / AMD Ryzen 5 or higher
- **Memory:** Minimum 16 GB RAM (32 GB recommended)
- **Storage:** At least 100 GB free SSD space
- **Graphics:** NVIDIA GTX 1650 / RTX Series or equivalent (recommended)
- **Display:** Full HD (1920 × 1080)





Requirements Specification

Software Development Requirements

- **Operating System:** Windows 10/11 or Ubuntu 22.04 LTS
- **Programming Languages:** C++, Python 3.10 or higher
- **GUI Framework:** Qt 6
- **Computer Vision:** OpenCV
- **Audio Processing:** Librosa
- **Video Processing:** FFmpeg
- **AI Model:** Small Language Model (SLM)
- **Development Tools:** Qt Creator / Visual Studio Code





Requirements Specification

- **IDE Used:** Visual Studio Code / Qt Creator
- **Version Control:** Git
- **Development Tools and Frameworks:**
 - Qt 6 – Desktop GUI development
 - Python 3.10+ – AI and backend processing
 - OpenCV – Video analysis and shaky footage detection
 - Librosa – Audio processing and silence detection
 - FFmpeg – Video trimming, rendering, and export
 - Small Language Model (SLM) / Ollama – Natural language prompt interpretation





Requirements Specification

Functional Requirements

- Accept multiple video clips as input.
- Interpret user editing instructions using natural language prompts.
- Detect shaky or low-quality video clips.
- Detect silence and remove unnecessary audio segments.
- Generate an AI-assisted rough cut automatically.
- Provide timeline-based Human-in-the-Loop editing.
- Allow users to accept, reject, or modify AI suggestions.
- Export the final edited video using FFmpeg.

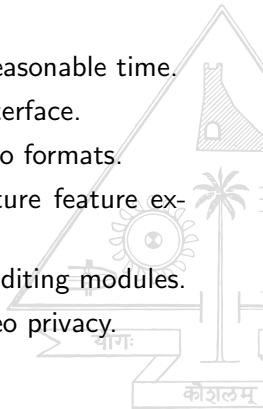




Requirements Specification

Non-Functional Requirements

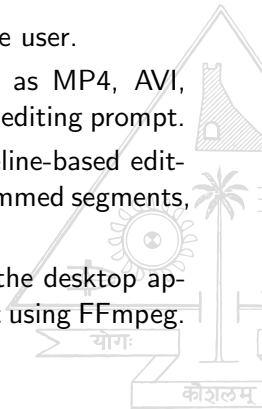
- **Performance:** Generate rough cuts within a reasonable time.
- **Usability:** Simple and user-friendly desktop interface.
- **Reliability:** Stable processing for different video formats.
- **Maintainability:** Modular architecture for future feature expansion.
- **Scalability:** Easy integration of additional AI editing modules.
- **Security:** Local processing to protect user video privacy.





Input and Output

- **Input:** Multiple raw video clips uploaded by the user.
- **Input Format:** Common video formats such as MP4, AVI, MOV, and MKV, along with a natural language editing prompt.
- **Output:** AI-assisted rough-cut video with timeline-based editing suggestions, detected shaky clips, silence-trimmed segments, and user-editable timeline.
- **Output Format:** Previewable timeline within the desktop application and final exported video in MP4 format using FFmpeg.





Conclusion

- SmartEdit combines Artificial Intelligence with a Human-in-the-Loop (HITL) workflow to simplify the rough-cut video editing process.
- The system uses natural language prompts, computer vision, and audio analysis to automatically detect shaky footage, remove silence, and generate an initial video edit.
- Unlike fully automated editors, SmartEdit allows users to review, modify, accept, or reject AI suggestions through an interactive timeline.
- The proposed framework reduces editing time while preserving creative control, making it suitable for content creators, YouTubers, wedding videographers, and multimedia professionals.



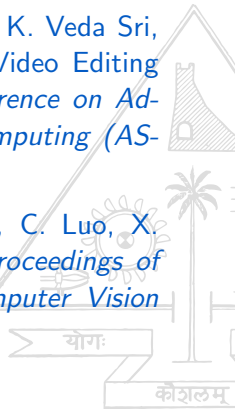
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- [1] T. Tilekbay, A. Kim, D. E. Park, and J. A. Landay, “ExpressEdit: A Multimodal System for Video Editing with Natural Language and Sketching,” *Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2024.
- [2] Z. Chen, K. Liang, H. Wang, *et al.*, “EditBoard: A Comprehensive Evaluation Benchmark for Text-based Video Editing,” *arXiv preprint arXiv:2402.09257*, 2025.
- [3] K. Wang, Z. Wang, Z. Yuan, *et al.*, “LAVE: An LLM-powered Agent for Video Editing,” *arXiv preprint arXiv:2402.19131*, 2024.



References

- [4] K. Shekar, K. Kasi Viswanadh, V. Srujan Varma, K. Veda Sri, S. Aditya, and S. Sahoo, "AI-Based Intelligent Video Editing System," *Proceedings of the International Conference on Advancements in Smart, Secure and Intelligent Computing (AS-SIC)*, 2025.
- [5] S. Tu, Q. Dai, Z. Zhang, S. Xie, Z.-Q. Cheng, C. Luo, X. Han, Z. Wu, and Y.-G. Jiang, "Paper Title," *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.





Thank You

